Daniel Arbeláez Álvarez

Software Engineer · University Lecturer

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Sources

PROFILE

Software engineer and university lecturer. Employee of the Judicial Branch of Colombia since 2014. In a personal capacity, I develop free software for operational tasks on electronic court case files, which does not weigh evidence or take part in judicial decisions. I am studying for the Specialization in Artificial Intelligence.

Technologies: Python (Streamlit, FastAPI, pandas, spaCy) · Language-model APIs (OpenAI, Anthropic) and RAG · Docassemble · Docker · PostgreSQL and SQLite · R · Power BI · Git and GitHub.

AgilEx

DNDA copyright registration (3 November 2022) and 2,426 downloads on GitHub. As «LAW», in the 2023 Legal Innovation Map by Legal Hackers Colombia and ALTCO. Sources: map.

JusticIALab 2024

Winner, Data-Driven Justice category, as a member of team Marduk. Mention of Recognition from the Superior Council of the Judiciary (Agreement PCSJA24-12237). Sources: CSJ · UTDI.

Cuaderno Activa

Two articles on LegalTech and artificial intelligence applied to justice: one published (2024) and one in press.

EXPERIENCE

Judicial Branch of Colombia · Employee
November 2014 – present
Since 2024, technology support and judicial information systems.

Tecnológico de Antioquia · Adjunct lecturer
February – November 2024

EDUCATION

Specialization in Artificial Intelligence
Fundación Universitaria Internacional de La Rioja (UNIR Colombia) · In progress, since 2025

Software Engineering
Tecnológico de Antioquia · 2021

LEGAL SOFTWARE DEVELOPMENT

AgilEx Since 2022 · MIT — Builds a case file's electronic index from file metadata; it does not assess content.

TYBABot 2025 · MIT — Guides users on Justicia XXI Web (TYBA). (template by another author)

DisciplinaJudicialAI 2026 · MIT — Guides users on the judicial disciplinary regime. (template by another author)

Docassemble Expert 2025 · MIT — Helps write and debug Docassemble interviews. (template by another author)

TutelaBot (adaptation) 2025 · MIT — Guides users on the tutela and the right of petition. (original project)

PUBLICATIONS AND TALKS

State of the Art of Symbolic AI Planners and Their Potential in LegalTech: Towards Neurosymbolic Architectures for Procedural Truth. Cuaderno Activa, in press (in Spanish).

Guided Interviews for the Optimization of Judicial Processes: an Approach from LegalTech. Cuaderno Activa, 16(1), 2024 (in Spanish). doi:10.53995/20278101.1858

Shared talk at the Academic Conference on AI and Its Impact on Justice (Medellín, 2024), in which I presented the Marduk case. (programme, in Spanish)

LANGUAGES

Spanish (native); English A2 (technical reading).

RECENT TRAINING

Artificial Intelligence in Judicial Decisions: Case-Law Analysis and Criteria for Applying the Law · Escuela Judicial Rodrigo Lara Bonilla · 2026 · 5 h

Fundamentals of Generative Artificial Intelligence (postgraduate level) · Escuela de Posgrado Newman · 2026 · 80 h

Microsoft .NET + Angular training cycle · Escuela Judicial Rodrigo Lara Bonilla e Intelligent Training · 2025 · 48 h

Artificial Intelligence Bootcamp, intermediate level · MinTIC – Talento Tech (Universidad de Antioquia y Universidad de Caldas) · 2025 · 159 h

In total, 5 diploma programmes (512 h) and 38 certified courses and activities since 2020.